

Hyperspectral imaging combined with convolutional neural network for outdoor detection of potato diseases

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Abstract—Potato is a kind of cultivated crops with a long history. It is the third most important food crop in the world, next to wheat and corn, and plays an important role in agricultural production. Potato late blight (PLB) is one of the main Potato diseases, which can cause 20–40% of the Potato yield loss and bring great losses to the Potato industry. Therefore, it is very important to identify the disease quickly and effectively and take correct control measures. In this study, a new outdoor classification method for potato late blight was proposed, and the outdoor data collected were divided into 5 categories. Due to the small number of experimental samples, a PLB early outdoor detection method combining convolutional neural network (CNN) and spectral data preprocessing method was proposed in this study. They are respectively CNN and multiple scattering correction (CNN-MS), CNN and wavelet transform (CNN-WT), CNN and first-order difference (CNN-D1), and CNN and second-order difference (CNN-D2). The experimental results show that the accuracy of CNN model test set reaches 97.18%, and the accuracy of CNN-WT and CNN-D2 model test set reaches 100%. The experiment proves the feasibility of using CNN and pretreatment method to detect PLB diseases under the new disease detection method. It provides a new direction for the detection and control of crop pests based on hyperspectral imaging technology.

Keywords—Hyperspectral imaging; Preprocessing; CNN; Disease Detection (key words)

I. INTRODUCTION

Crop diseases are caused by various factors in the process of planting, if the effective measures are not taken to prevent and control, they will affect the yield and quality seriously. Disease detection is important to ensure the healthy growth of crops. At present, the diagnosis of crop disease relies mainly on, field observation, pictures, text by plant protection experts who have the professional knowledge, which can be time-consuming and requires high level of professional technology, in consequence, the crop production may be reduced for not diagnosing and controlling in time[1]. Hyperspectral imaging technology is a new technology which combines traditional imaging technology and spectral technology effectively[2], which integrates image information and spectral information. CNN is a deep learning method emerging in recent years, which has a strong learning ability and has been widely used in the fields of computer vision and speech recognition for multi-classification and large-scale data and achieved great success[3–5]. At present, hyperspectral imaging technology combined with convolutional neural network algorithm is seldom used in crop disease diagnosis, especially in outdoor disease detection. Therefore, outdoor

potato was taken as the research object in this study, hyperspectral imaging technology was adopted to obtain sample hyperspectral data, and multiple scattering correction[6], wavelet transform[7], first-order difference[8] and second-order difference[9] were used to obtain average spectral reflectivity data for spectral preprocessing by selecting regions of interest in sample collection. By establishing models of CNN, KNN, back propagation(BP) and different pretreatment method combining potato disease detection quantitative analysis of different discriminant accuracy of the model is achieved, which provides theoretical and technical support for disease detection of outdoor potatoes.

II. EXPERIMENT PART

A. Experimental material acquisition and data acquisition

The Sample of potato and potato late blight pathogen are provided by Institute of Potato of Yunnan Normal University. The selection of experimental samples and inoculation of the pathogen were conducted under the guidance of experienced researchers from the Potato Research Institute. 10 healthy potato plants are selected randomly in Potato Planting Base in Yunnan Normal University and 5 of them were inoculated with *Phytophthora infestans*. The healthy and inoculated potato plants were cultured at a certain distance to prevent the healthy plants from being infected by the inoculated plants and the plants were cultured for 6 days. On the 6th day, 10 plants were put together in the potato planting base, and the visible light hyperspectral camera was used for data acquisition. Acquisition process is shown in figure 1, the left is the portable hyperspectral cameras, the right is the visualized results selected from wavelengths of visible light, 35, 17 to 53 which collected by ENVIS.1 software for hyperspectral image RGB. As you can see from the right picture of figure 1, hyperspectral image data contains five types-- healthy potato, potato late blight, floor, clay, and flower pot. Therefore this study used ROI ENVIS.1 software tools to select no less than 100 samples of each type of samples, and extracted spectral builder of average spectral reflectance by using pop processing tools of the spectral library software.

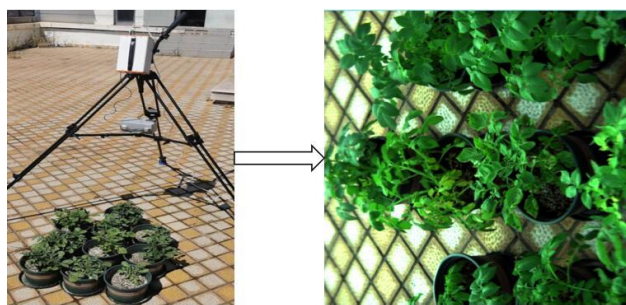


Figure 1. Hyperspectral imaging system.

B. Spectral pretreatment

Average spectral reflectance data of the region of interest (ROI) of 5 types of samples obtained through ENVIS.1 were plotted using the Origin 2017 drawing tool, as shown in Figure 2(a). In order to find out the similarities and differences of spectral data of each type of sample, the average value of each type of sample data was calculated, as shown in Figure 2(b).

Spectral analysis is widely used in remote sensing, medicine, agriculture, industry and other fields because of its advantages of sensitivity, rapidness and easy operation. Practical application is shown in Figure 2(a) in, spectral data has multiple data dimensions, a large amount of data redundancy and is easily affected by external environment, noise, baseline drift and other factors, resulting in noise. As can be seen from Figure 2(b), the spectral curves of outdoor floor, outdoor soil and outdoor flower pot show the same trend, and the curves of potato health and potato late blight show the same trend. However, the reflectivity values of different classes show significant differences in different bands. The spectral curves of healthy potatoes and infected potatoes have almost the same fluctuation trend. There are obvious reflectance peaks around 540nm and 750nm, and there are obvious absorption peaks around 680nm, and then the reflectance suddenly rises sharply. The spectral reflectance of infected and healthy potatoes was significantly different in visible and near infrared wavelengths. These differences are caused by changes in leaf internal element content, damage of cell tissue, loss of chlorophyll and water during the occurrence of disease, which makes the reflectivity of healthy leaves lower than that of disinfected leaves in the visible light band, while it is higher than that of disinfected leaves in the near infrared band.

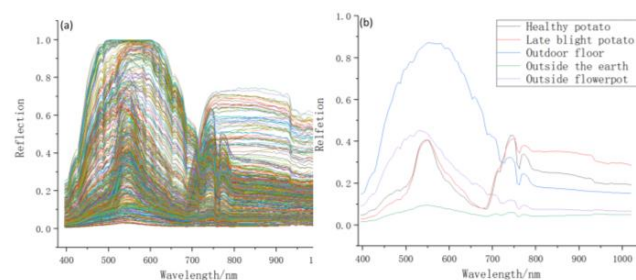


Figure 2. Spectral chart.

Before the disease detection of outdoor potato, the appropriate pretreatment method is selected to preprocess the spectral data of the samples, which can affect the experimental results in different degrees. At present, there are many pretreatment methods for spectral data, and the appropriate pretreatment method needs to be determined by experimental results. In this study, MSC, WT, D1, D2 were used for data preprocessing. D1, D2 and WT can eliminate the influence of instrument background or drift on spectral signal. MSC was used to eliminate the scattering phenomenon caused by uneven particle distribution and particle size difference and the influence of optical path transformation on diffuse reflection. Figure 3. shows the spectral data of potato sample leaves after using the above 4 pretreatments. It can be seen from Figure 3(a) that MSC eliminates the influence of scattered color on the spectrum. From (b-d), we can also find that the spectrum processed by WT, D1 and D2 obviously solves the irrelevant information such as baseline drift and sample background.

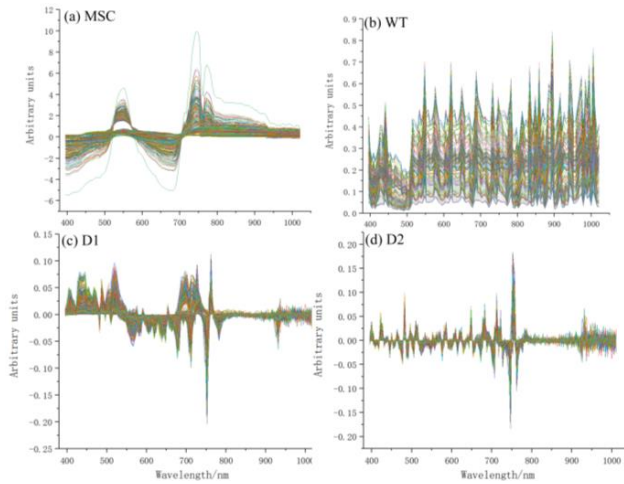


Figure 3. Spectral data preprocessing.

III. POTATO DISEASE DETECTION MODEL

A. CNN

CNN is an artificial neural network designed by simulating biological neural network. Weight sharing reduces the complexity of CNN, especially the feature that images of multi-dimensional input vectors can be directly input into the network avoids the complexity of data reconstruction in the process of feature extraction and classification. Compared with the traditional deep learning algorithm, it directly processes the data and extracts the features automatically without selecting the features manually or using any feature extraction algorithm. It learns the corresponding features automatically in the training and learning process to realize pattern classification. Spectral data and image data have different data dimensions, so the CNN adopted is one-dimensional CNN. Building a CNN model is often the most essential step. The model extracts features through a convolutional layer, the last layer of the network for each class in the classification task. What the classification algorithm obtains is a decision surface. By comparing the probability of spectral reflectance data of each band belonging to each category, it can judge which category it belongs to. After several parameter adjustment attempts, 7-layer Conv1D was finally used to extract the characteristic values, and a layer of MaxPooling1D was added after every two layers of Conv1D to retain the main features. Choosing a pool layer of size 3 means that the size of the output matrix of this layer is only one third of the size of the input matrix, which can reduce the computational effort. The hyperbolic tangent function (tanh) was used in each convolutional layer to improve the expression ability of the neural network to the model. tanh is often used in multi-classification tasks as an activation function. The neural network is not interpretable, and the number of convolution kernels and fully connected nodes is calculated by an exponential power of 2. The Flatten () layer acts as an intermediate layer to link the Convolutional Neural Network to the fully connected layer. The neural network is trained to minimize a loss function (Softmax loss) that captures the difference between the characteristic output of the network and a given expected output. Five output nodes correspond to five categories. Softmax matches each feature data to the

feature category with the highest probability. As the loss function of model training, the categorical crossentropy describes the distance between the probability distribution learned at present and the actual probability distribution. The smaller the loss function is, the more similar the two probability distributions are. The CNN model constructed in this paper is shown in Figure 4.

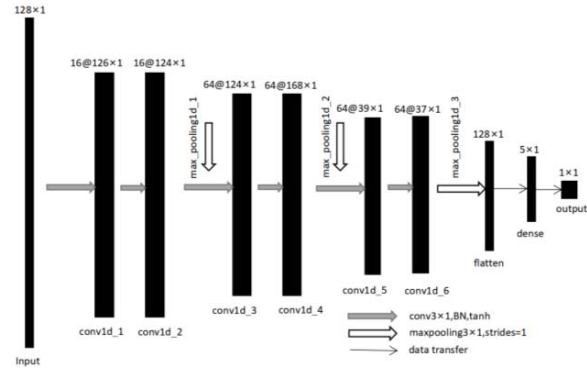


Figure 4. CNN model structure.

In order to reflect the stability and rationality of the CNN model, the accuracy and the change process of the loss curve in the process of model training and verification are visualized. Figure5 shows the change curve of the accuracy and loss values of the training set and the verification set when the CNN model constructed is applied to the recognition process of the spectral data set without any preprocessing. It is found that the accuracy curve and loss curve of training set and verification set almost coincide. After 200 iterations of the 5 kinds of outdoor spectral data collection, the accuracy of training set and verification set showed a linear upward trend before 25 iterations, and then leveled off, reaching 97.18% in the end. In the process of training and verification, there are errors and jitter in the precision and loss curves, but as the number of iterations increases, the accuracy continues to improve, and the loss value continues to decrease, and finally flattens out. It is concluded that the CNN model constructed has fast convergence speed and strong generalization ability.

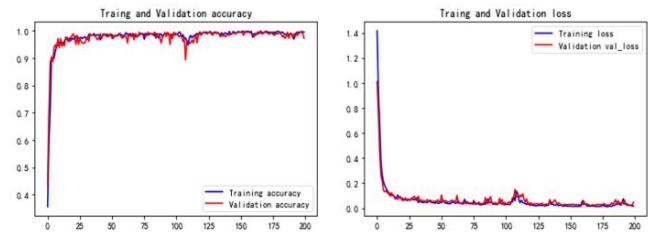


Figure 5. The change curve of CNN accuracy and loss value based on the original spectral information.

B. Introduction to other comparative experimental algorithms

The comparative experimental algorithms selected in this study include KNN and BP. The two methods are described as follows.

KNN calls the KNeighborsClassifier library directly. Parameters call the GridSearchCV in the Python learning library to select the best parameters. After training and learning, the KNN model input the test set data without labels, compared each feature in the sample data of the test set with the corresponding feature in the sample data of the training set, and extracted the classification label of the data with the most similar features in the sample set. Select the first k most similar data in the sample data of the training set, and finally select the classification that appears the most times in the k most similar data as the classification of the sample data of the test set.

BP adjusts the weight of the hidden layer indirectly by the error between the output result obtained by the output layer and the expected output. A multi-layer feedforward network trained by error back propagation algorithm is one of the most commonly used neural network models.

IV. RESULTS AND ANALYSIS

The sample data is divided into 70% training set and 30% test set. After training with the training set, the test set is input for identification, and the detection results and confusion matrix can be obtained. Different spectral processing methods and different machine learning models were combined to compare their accuracy and record them in Table 1. In Table 1, the comparison between the results of different spectral data processing methods and the original spectral data is firstly observed horizontally. It can be seen that all of the four spectral data pretreatment methods selected in this study can improve the identification accuracy in various degrees. Therefore, in the case of specific model performance, in order to improve the detection effect, we can improve the experimental results by combining preprocessing method and machine learning method. The combination of different machine learning methods and different preprocessing methods will have different effects. CNN+WT and CNN+D2 improve the accuracy from 97.18% to 100%, and KNN+D2 from 95.76% to 100%. It is very important to get the reasonable use of the pretreatment method. Through longitudinal comparison of different machine learning models, CNN results are the best for spectral data processed by the same pretreatment method, and its identification accuracy is higher than the other two machine learning methods. The machine learning model based on CNN has a high precision of recognition after MSC, WT, D2 and D1 processing, with the lowest being 99.44%. The accuracy of KNN and BP after MSC, WT, D2 and D1 processing is significantly different from that of CNN. It is concluded that convolutional neural network combined with hyperspectral imaging technology is feasible to apply to potato disease detection, and has a better application prospect than traditional machine learning methods such as BP and KNN.

TABLE I. DIFFERENT PROCESSING METHODS WERE USED TO COMPARE THE PERFORMANCE OF THE TEST SET IN DIFFERENT MODELS(ACCURAY%).

Type	MSC	WT	D1	D2	RAW
CNN	99.44	100	100	99.44	97.18
KNN	97.23	96.81	100	98.78	95.76
BP	95.16	93.85	97.16	95.47	89.56

Although the overall classification accuracy can reflect the overall classification effect of multiple classification models, it cannot reflect the mutual interference between different kinds of spectral information. Therefore, it is necessary to analyze the classification effect of multi-classification model for outdoor potato disease detection by confusion matrix. Figure 6 shows the confusion matrix of the discriminant learning detection results of the spectral data with different processing methods combined with the convolutional neural network, where RAW is the confusion matrix for the detection of the original data. All the correct classifications in the confusion matrix diagram are concentrated on the diagonal, and the closer the accuracy to 1 on the diagonal is, the better the classification effect of the model is. Taking RAW confusion matrix as an example, the horizontal axis is the predicted category and the vertical axis is the true category. From the figure, we can find that the recognition rate of health, outdoor floor and outdoor soil is 100%, among which, 7% of late blight disease was wrongly judged as healthy, and 14% of outdoor flowerpot was wrongly judged as outdoor floor. By comparing the confusion matrix results of different treatment methods, it can be seen that the best judgment results for health, outdoor soil and outdoor floor are 100%. Under the classification model of the original data, 9% of late blight diseases were misdiagnosed as healthy, and 5% of late blight diseases were misdiagnosed as healthy under the first-order difference. As a result, the difference between the disease spectrum data and the healthy spectrum data is not obvious and easy to be misclassified, indicating that there is mutual interference between the health and the late blight disease, and the performance is not stable enough.

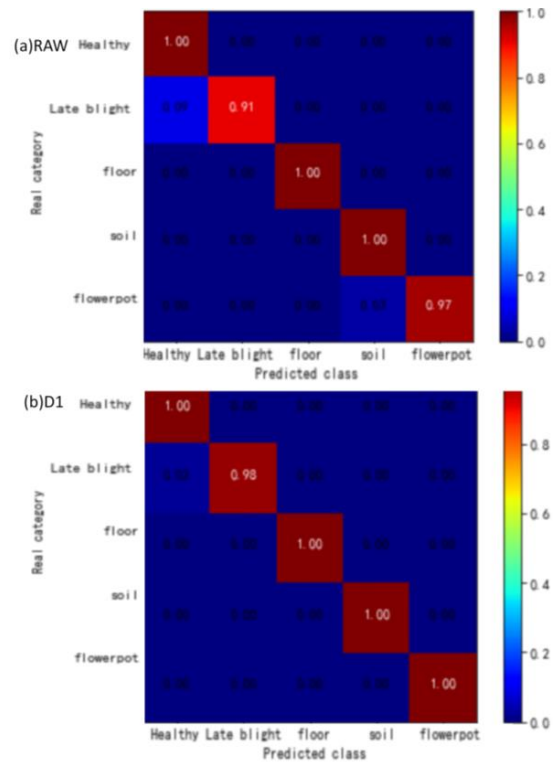


Figure 6. CNN identification results confusion matrix.

V. CONCLUSION

In view of the existing research, hyperspectral imaging technology combined with deep learning method is seldom applied to outdoor crop disease detection, and the efficiency of classification of traditional machine learning method is not good, and spectral reflectivity data has problems like spectral peak overlap, so spectral preprocessing method was introduced to preprocess the original spectral data. Research on detection of outdoor potato diseases was carried out. A simple but very effective one-dimensional deep convolutional neural network model was designed, and the effects of four spectral data preprocessing methods on outdoor potato disease diagnosis were analyzed. Compared with the traditional methods KNN and BP, CNN has achieved higher accuracy of classification and good scalability in outdoor crop disease detection. Combined with WT and D2 pretreatment methods, the accuracy of CNN model reached 100%. In conclusion, it is feasible to combine convolutional neural network and hyperspectral technology to detect crop diseases and insect pests in the field environment, which provides effective support for the detection and control of crop diseases and insect pests.

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